



Bank Loan Applications

Data Cleaning & Exploratory Analysis using Python

Project Overview

This project demonstrates Python, Pandas, and Matplotlib skills by cleaning and analyzing a realistic dataset of 25,000 bank loan applications.



Data Loading & Auditing

Initial assessment of the raw dataset.



Column-by-Column Cleaning

Systematic cleaning of each data field.



Feature Engineering

Calculating EMI for deeper insights.



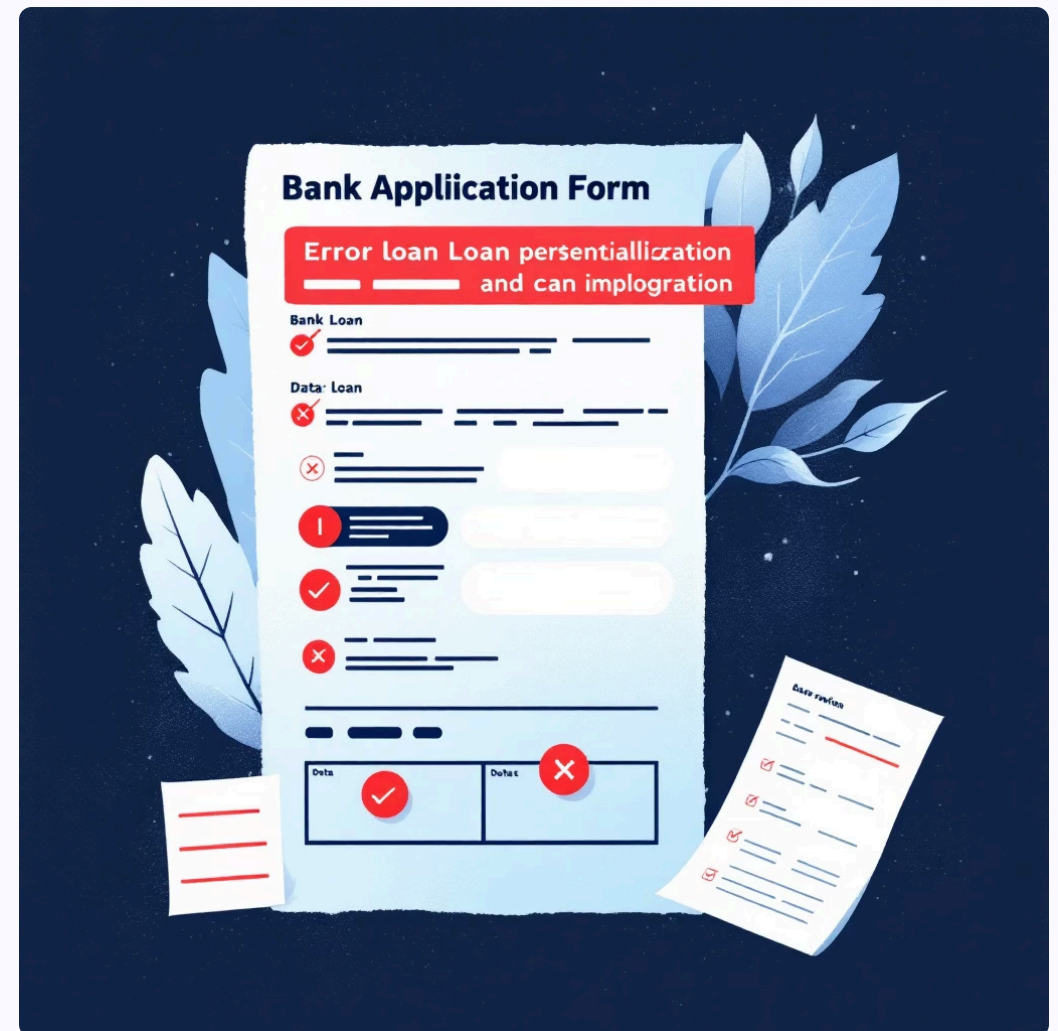
Clear Visualizations

Presenting insights effectively.

Dataset Description: 13 Columns, Many Inconsistencies

The dataset includes critical information for loan applications, each with specific data quality issues:

- Application_ID: Missing values
- Gender: Mixed cases, spaces
- Age: Invalid ages (<18, >100)
- Education: Mixed cases
- Marital_Status: Mixed cases
- Employment_Type: Inconsistent formats
- Monthly_Income: Commas, strings, missing



- Loan_Amount: Commas, strings
- Loan_Term: Missing values
- Credit_Score: Invalid values (<300, >850)
- Existing_Debt: Comma values
- Approval_Status: Inconsistent formatting
- Application_Date: Invalid or missing dates

Identified Data Problems

Categorical Issues

- Mixed-case categories
- Improper spacing
- Incorrect labels
- Missing values

Date & Missing Values

- Invalid or missing dates
- 3% missing values in every column

Numeric Issues

- Commas in numbers
- String-based numbers
- Out-of-range values
- Invalid credit scores



Data Cleaning Steps: A Systematic Approach



Standardize Text

Converted all strings to proper case and cleaned categorical values.



Clean Numeric Fields

Removed commas, converted to float/int, and removed unrealistic values.



Fill Missing Values

Used median for numeric, mode for categorical, and specific logic for IDs/dates.



Feature Engineering

Calculated EMI using a standard loan formula.

Feature Engineering: EMI Calculation

Equated Monthly Installment (EMI) was calculated to provide a comprehensive view of loan affordability.

$$EMI = P \times r \frac{(1 + r)^n}{(1 + r)^n - 1}$$

Where:

- **P** = Loan amount
- **r** = Monthly interest rate (10% annual / 12)
- **n** = Loan term in months



Visualizations & Key Insights



Monthly Income

Most applicants earn ₹20,000-₹60,000, with few high-income outliers.



Loan Amount

Majority of loans are ₹200,000-₹600,000, with some high-value loans.



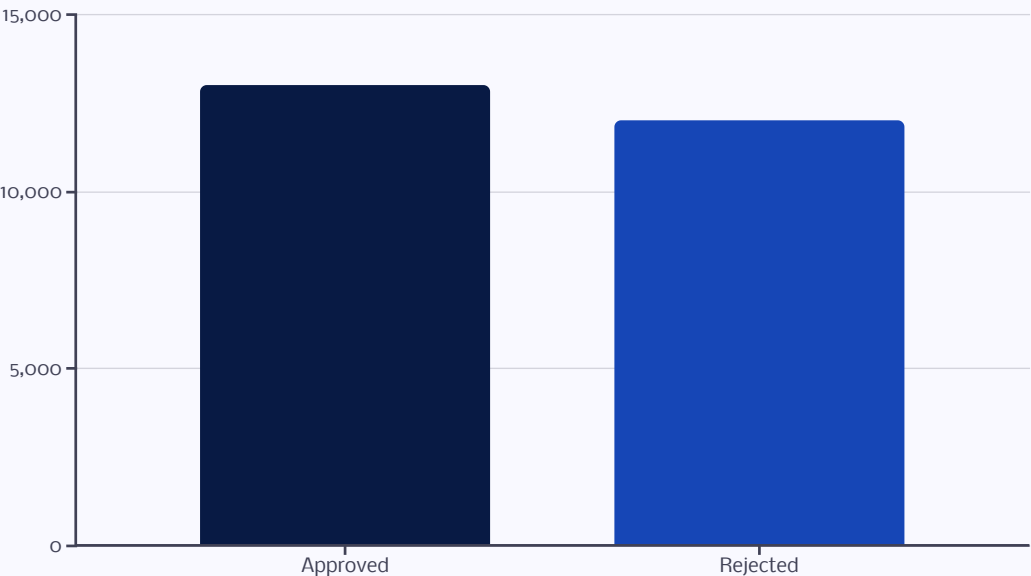
Credit Score

Scores mostly between 550-750, outliers successfully cleaned.



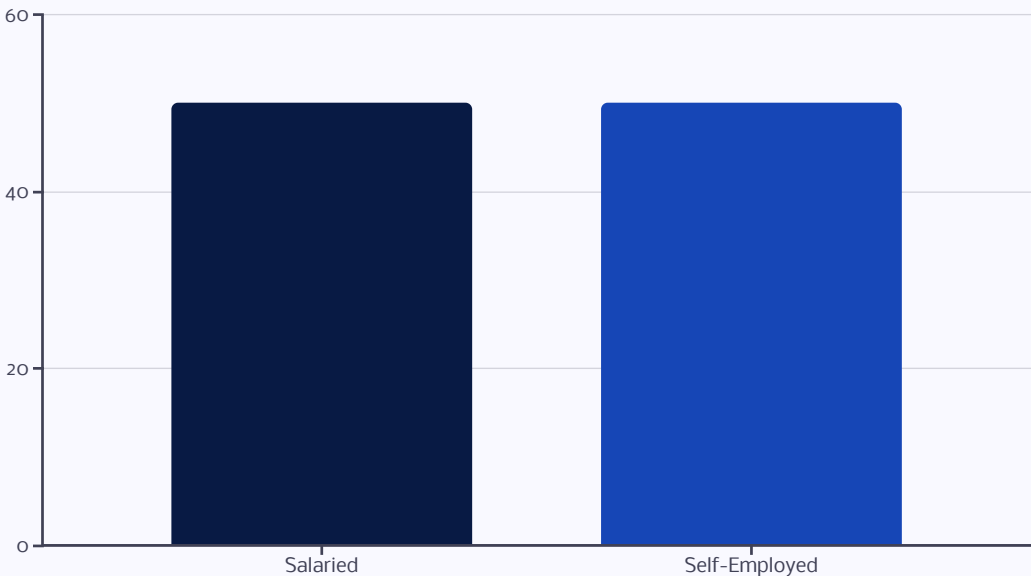
Approval Status & Employment Type

Loan Approval Status Count



The bar chart illustrates the count of loan applications by status, showing approximately 13,000 applications were Approved and 12,000 were Rejected. This indicates a relatively balanced outcome between approvals and rejections.

Approval Rate by Employment Type



The bar chart shows that both salaried and self-employed individuals have an approximate loan approval rate of 50%. This suggests that the approval process evaluates both employment types with similar outcomes.

Final Cleaned Dataset

The rigorous cleaning process resulted in a high-quality dataset, ready for further analysis.

Zero Missing Values

All columns are now complete.

Standardized Categories

Consistent and clean categorical data.

Valid Numeric Fields

Accurate and within-range numerical data.

Proper Date Formats

All dates are correctly formatted.

New EMI Column

Enhanced with calculated EMI values.

Project Demonstrates Core Data Skills

This project showcases a comprehensive skill set in data handling and analysis.



Messy Data Handling

Proficiency with real-world, imperfect data.



Cleaning & Validation

Expertise in managing missing and invalid values.



Data Transformation

Ability to convert data types and engineer features.



Insightful Visualizations

Creating clear and impactful data representations.